

Addressing Stock-Turnover Disparity at Kirtan Hardware: An Inventory and Sales Optimization Approach

A Mid-Term report for the BDM capstone Project

Submitted by

Name: Anirudh Pabbaraju

Roll number: 23f2002166



IITM Online BS Degree Program,
Indian Institute of Technology, Madras, Chennai
Tamil Nadu, India, 600036

Contents :

1) Executive Summary	3-4
2) Proof of Originality of Data	4-5
3) Metadata and Descriptive Statistics	5-9
3.1) Metadata	5-7
3.2) Descriptive Statistics	7-9
4) Detailed Explanation of Analysis Process/Method	9-10
5) Results and Findings	10-12

Declaration Statement

I am working on a Project titled “**Addressing Stock-Turnover Disparity at Kirtan Hardware: An Inventory and Sales Optimization Approach**”. I extend my appreciation to **Kirtan Hardware, Hyderabad, Telangana**, for providing the necessary resources that enabled me to conduct my project.

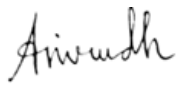
I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

Signature of Candidate: 

Name: Anirudh Pabbaraju

Date: 1/04/2025

1) Executive Summary

This document addresses stock turnover disparity at kirtan hardware and what have been done in subsequent below sections

1.1) Business Context & Problem

Kirtan Hardware, a major wholesale distributor in South India, has scaled rapidly to 180 outlets. However, a severe stock-turnover disparity exists—₹1 crore in inventory yields only ₹45 lakh in sales—causing unsold stock, blocked capital, and declining profitability.

1.2) Proof of Originality

Study authenticity confirmed via an official letter and owner-recorded video

1.3) Data Sources & Metadata

Analysis spans five validated datasets: stock_data, sales_data, finance_data, operation_data, and customer_feedback_data, with metadata review and standardization applied across files.

1.4) Descriptive Statistics & EDA

- * "Knobs" are the most overstocked items.
- * “Revenue is right-skewed, peaking around ₹30,000.
- * "Kadi" shows highest employee efficiency; "Handles" underperform (<80%).
- * Applied EDA techniques: histograms, box plots, correlations, and preprocessing.

1.5) Key Results & Findings

- * Break-even points cluster near profitability, but outliers exist.
- * Most products exceed the break-even line; “Accessories” show consistent margins
- * Demand forecasts and stock trends reveal allocation mismatches across categories.

1.6) Outcome & Impact

Techniques like Inventory Optimization, Sales Forecasting, Profitability & Risk Analysis, and Efficiency Assessment provide actionable insights to reduce inefficiencies, optimize stock levels, and boost profitability.

2) Proof of originality of the Data-

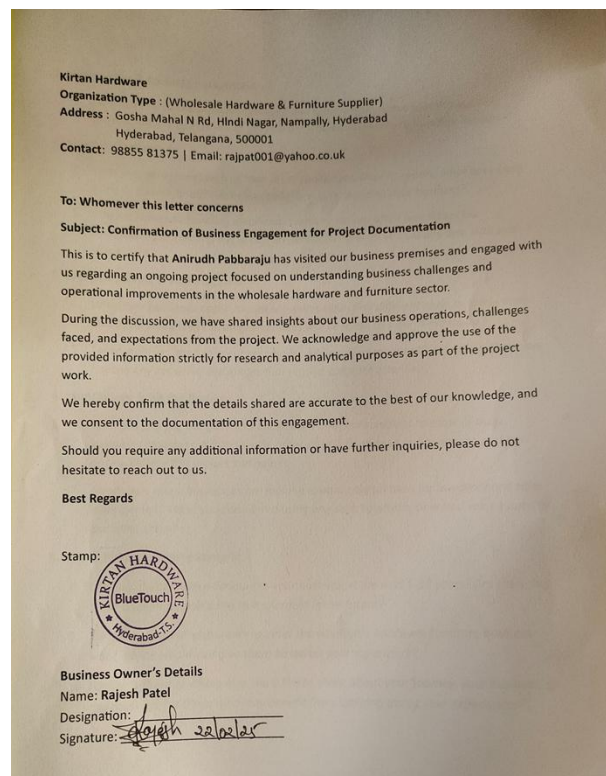


Figure-1 Letter of authenticity of data



Figure-2 SHOP-PIC (FIRST MEET WITH OWNER)



Figure-3 SHOP-NAME

Link containing more photos of the KIRTAN HARDWARE SHOP-

https://drive.google.com/drive/folders/1IUVVh93uEE5QwVMcJTahtlS9iy_Qa1zI?usp=sharing

Repository containing Data, Survey and Data Wrangling-

<https://github.com/Anirudh-starhash/BDM-PROJECT>

Google Drive Link of Video Interaction with the Founder-

<https://drive.google.com/file/d/1mu4OKskwcOiMPiQ1vCWtWaA9MMCEcg2l/view?usp=sharing>

3) Metadata and Descriptive Statistics

3.1) Metadata:

3.1.1) Metadata of Stock_dataP.csv:

Column Name	Type	Info & Potential Use
Stock_ID	(int)	Unique identifier for stock items. Used for tracking.
Item_Name	(str)	Name of the stock item. Helps in product-level analysis.
Category	(str)	Classification of items. Used in inventory segmentation.
Quantity_Available	(int)	Current stock level. Crucial for stock monitoring.
Restock_Threshold	(int)	Minimum stock level before reorder. Helps avoid stockouts.
Purchase_Cost	(float)	Cost per unit of item. Used in profitability analysis.
Selling_Price	(float)	Price per unit for sales. Used for revenue calculations.
Supplier	(str)	Source of the stock. Used for supplier performance tracking.
Defective_Stock	(int)	Number of defective units. Helps in quality control.
Revenue	(float)	Revenue generated per item. Used for financial insights.

3.1.2) Metadata of Sales_dataP.csv:

Sale_ID	(int)	Unique identifier for each sale. Helps track transactions.
Stock_ID	(int)	Links sale to stock item. Used for inventory correlation.
Item_Name	(str)	Name of the sold item. Helps in product performance analysis.
Category	(str)	Classification of item. Used for segment-wise sales analysis.
Sale_Quantity	(int)	Number of units sold. Helps in demand forecasting.
Sale_Price	(float)	Price per unit at the time of sale. Used for revenue analysis.
Discount	(float)	Discount applied on sale. Helps in pricing strategy analysis.
Customer_Region	(str)	Region where the sale occurred. Helps in geographic trends.
Return_Status	(bool)	Indicates if the item was returned. Used for return rate analysis.
Total_Revenue	(float)	Revenue from each sale. Used for overall financial performance.

3.1.3) Metadata of Finance_data.csv:

Column Name	Type	Info & Potential Use
Finance_ID	(int)	Unique identifier for financial records. Helps track financial transactions.
Stock_ID	(int)	Links finance data to stock items. Used for financial inventory analysis.
Item_Name	(str)	Name of the item. Helps in profitability assessment per product.
Category	(str)	Classification of item. Used for category-wise financial insights.
Date	(datetime)	Date of financial entry. Helps in time-series financial analysis.
Revenue	(float)	Total income from item sales. Used to calculate financial performance.
Cost	(float)	Cost of goods sold (COGS). Helps determine gross profit.
Profit	(float)	Revenue - Cost. Key metric for business success.
Tax	(float)	Tax paid on revenue. Helps in net profit calculations.
Net_Profit	(float)	Profit after tax. Used for business profitability assessment.

3.1.4) Metadata of Operation_data.csv:

Column Name	Type	Info & Potential Use
Operation_ID	(int)	Unique identifier for each operational record. Helps track efficiency trends.
Stock_ID	(int)	Links operational data to stock items. Used for analyzing operational impact on inventory.
Item_Name	(str)	Name of the item. Useful for identifying bottlenecks in specific products.
Category	(str)	Classification of item. Helps analyze performance by product type.
Date	(datetime)	Date of operation. Essential for time-based trend analysis.
Orders_Processed	(int)	Number of orders completed. Key metric for operational efficiency.
Employee_Efficiency_Percentage	(float)	Efficiency of employees in percentage. Helps in workforce performance evaluation.
Machine_Downtime_Hours	(float)	Hours machines were non-operational. Critical for identifying production bottlenecks.
Delayed_Orders	(int)	Number of late orders. Helps in analyzing process inefficiencies.
Returned_Orders	(int)	Number of returned orders. Useful for identifying quality/supply chain issues.

3.1.5) Metadata of Customer_feedback.csv:

Column Name	Type	Info & Potential Use
Feedback_ID	int	Unique identifier for each feedback entry. Useful for tracking and referencing specific feedback.
Customer_Name	str	Name of the customer providing the feedback. Helps in personalizing responses and analyzing feedback trends among different customer segments.
Stock_ID	int	Identifier linking feedback to specific stock items. Facilitates analysis of customer sentiments related to particular products.
Item_Name	str	Name of the item. Assists in identifying which products are receiving positive or negative feedback.
Category	str	Classification of the item. Enables category-level analysis of customer satisfaction and product performance.
Date	datetime	Date when the feedback was provided. Essential for analyzing trends and patterns over time.
Rating	int	Customer's rating of the product, typically on a predefined scale (e.g., 1 to 5). Useful for quantitative analysis of customer satisfaction.
Feedback_Comment	str	Detailed comments provided by the customer. Valuable for qualitative analysis and understanding specific issues or praises mentioned by customers.

3.1.6) SOURCE CODE FOR BDM DATA IS IN THIS REPO-

https://github.com/Anirudh-starhash/BDM_DATA_CODES

3.2) DESCRIPTIVE STATISTICS

Descriptive statistics of Stock_data.csv

	Quantity_Available	Restock_Threshold	Purchase_Cost	Selling_Price	Defective_Stock	Revenue	Cumulative_Revenue	Cumulative_Percentage	COGS	Stock_Turnover
count	5000.000000	5000.000000	5000.000000	5000.000000	5000.000000	5.000000e+03	5.000000e+03	5000.000000	5.000000e+03	5000.000000
mean	491.670241	33.708095	537.966759	916.824124	5.805536	4.528988e+05	1.709412e+09	75.487604	2.659759e+05	538.187622
std	362.264924	31.014342	283.362920	492.238463	4.300096	4.475445e+05	5.968929e+08	26.358778	2.607434e+05	282.349549
min	0.000000	5.000000	50.000000	72.380740	0.000000	0.000000e+00	3.882796e+06	0.171464	0.000000e+00	50.000000
25%	183.000000	10.000000	296.750000	504.222704	3.000000	1.127192e+05	1.360109e+09	60.062364	6.638700e+04	299.000000
50%	461.000000	20.000000	547.000000	922.500509	5.000000	3.072944e+05	1.949424e+09	86.086530	1.842750e+05	545.500000
75%	745.000000	50.000000	765.000000	1293.042710	8.000000	6.772696e+05	2.200533e+09	97.175491	3.985835e+05	765.000000
max	1588.000000	100.000000	1467.375000	2476.272718	15.500000	3.882796e+06	2.264494e+09	100.000000	2.300844e+06	1467.375000

The stock quantity and revenue distributions are right-skewed, indicating that most items have low to moderate values, while a few drive the majority of stock and sales. Knobs are the most stocked among key categories like Handles, Kadi, and Accessories. These patterns highlight stock imbalances and underperformance, guiding smarter procurement and improving turnover efficiency.

Descriptive statistics of Sales_data.csv

	Sale_Price	Discount	Total_Revenue	Cumulative_Revenue	Cumulative_Percentage
count	5000.000000	5000.000000	5000.000000	5.000000e+03	5000.000000
mean	1745.656972	11.352756	41698.754929	1.474515e+08	70.722274
std	721.065959	7.442160	31259.322867	5.806625e+07	27.850354
min	500.390000	0.000000	0.000000	1.216856e+05	0.058364
25%	1123.172500	5.000000	16726.460000	1.085013e+08	52.040556
50%	1745.656972	10.000000	35988.050000	1.658100e+08	79.527544
75%	2383.045000	15.000000	58710.120000	1.983515e+08	95.135476
max	2999.510000	30.000000	121685.610000	2.084938e+08	100.000000

Sales data shows most transactions occur in low to mid quantities, with revenue peaking at key intervals. Knobs and Handles lead in sales, followed by Kadi and Accessories.

Regionally, the East dominates, with Central and South close behind. These insights support targeted inventory and marketing strategies for improved sales performance.

Descriptive statistics of Finance_data.csv

	Cost	Profit	Tax	Net_Profit
count	4806.000000	4806.000000	4806.000000	4806.000000
mean	11168.385178	13489.183964	2603.349265	10788.732016
std	5243.991346	7027.224947	1506.168238	5622.370138
min	2063.100000	-7088.775000	0.000000	-5671.026250
25%	6831.072500	8018.407500	1444.097500	6414.725000
50%	11101.500000	13069.935000	2457.115000	10455.950000
75%	15074.765000	17620.185000	3524.035000	14096.142500
max	27611.911250	32087.825000	6653.228750	25670.263750

Profitability analysis shows all categories maintain positive margins, with "Handles" having the highest median. While a few outliers indicate extreme profits or losses, most transactions stay above the break-even

point. Both revenue and profit are right-skewed—revenue peaks near ₹30,000 and profit centers around ₹15,000—indicating overall healthy performance despite occasional losses.

Descriptive statistics of Operation_data.csv

	Orders_Processed	Employee_Efficiency_Percentage	Machine_Downtime_Hours	Delayed_Orders	Returned_Orders
count	5000.000000	5000.000000	5000.000000	5000.000000	5000.000000
mean	131.383260	78.073672	10.560346	10.230400	4.95420
std	69.451395	9.872607	5.634236	6.234107	3.41361
min	0.000000	51.850000	1.400000	0.000000	0.00000
25%	73.000000	71.500000	6.100000	5.000000	2.00000
50%	131.000000	77.700000	9.900000	9.000000	4.00000
75%	180.600000	84.600000	13.800000	14.000000	7.00000
max	342.000000	104.250000	25.350000	27.500000	14.50000

Operation data reveals that "Kadi" has the highest employee efficiency, while

"Handles" falls below the 80% benchmark. Most days see 50–200 orders processed, showing a right-skewed trend. Increased machine downtime correlates with order delays, with a median downtime of 10 hours. These findings highlight the need to boost workforce efficiency and reduce downtime for better operations.

4) Detailed Explanation of Analysis Process/Methods

4.1) Data Collection :

The project draws data from five core CSV files—**stock_data.csv**, **sales_data.csv**, **finance_data.csv**, **operation_data.csv**, and **customer_feedback.csv**. Each file represents a distinct business domain, covering inventory, sales, financials, operations, and customer sentiment respectively.

4.2) Data Cleaning & Preprocessing :

The raw datasets contained inconsistencies, missing values, and formatting issues. These were addressed using:

- **Missing Values:** Identified using `df.isnull().sum()`; imputed with mean for numerics and mode for categoricals.
- **Outlier Treatment:** Capped using the IQR method to reduce skew. **Data Type Consistency:** Ensured correct data types for analysis.
- **Standardization:** Applied `str.lower()` and `str.strip()` to clean ID fields like `Stock_ID`, `Sale_ID`, etc. **Merge Preparation:** Cleaned datasets were merged using `pd.merge()` on common keys with an inner join to retain only relevant matched records.

4.3. Exploratory Data Analysis (EDA)

Once cleaned, EDA was performed using:

- `df.describe()` for statistical summaries
- **Univariate & Bivariate Analysis:** Correlation matrices for numerical dependencies, scatter plots for relational insights, and box plots to validate outlier treatment
- **Distribution Analysis:** Used `sns.histplot()` and `countplot()` from Seaborn to visualize patterns such as:
 - Quantity vs Revenue in stock data
 - Total Revenue in sales
 - Break-even points in finance
 - Orders processed and machine downtimes in operations

4.4. Analytical Methods by Domain

To extract business-specific insights, five tailored methods were applied:

- **Inventory Optimization** (*stock_data.csv*): Identified overstocked items and poor turnover patterns using ABC classification and turnover ratios.
- **Sales Prediction & Trend Analysis** (*sales_data.csv*): Modeled revenue patterns and regional trends through time series analysis and decomposition.
- **Profitability & Risk Assessment** (*finance_data.csv*): Assessed margins and breakeven points to evaluate financial sustainability and detect anomalies. And BEP vs Revenue was also calculated
- **Efficiency Optimization** (*operation_data.csv*): Analyzed employee performance, order volume, and downtime impacts to improve process flow

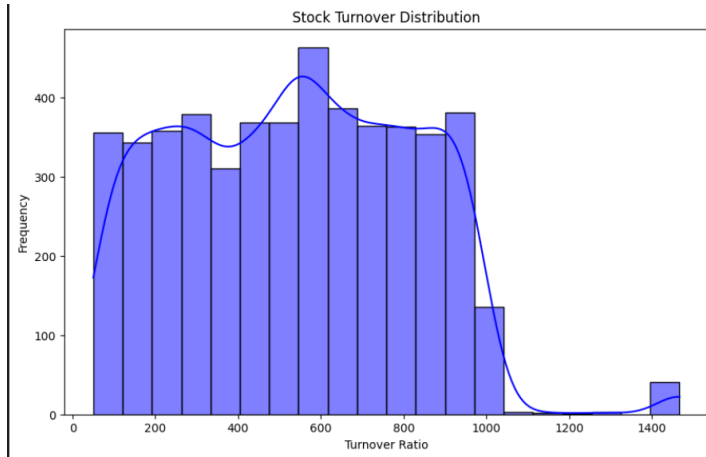
5) Results and Findings-

After Categorizing `stock_data` and `sales_data` into A,B,C types the counts are....

A: 2109, C: 1619, B: 1272 [for Stock Data] , A : 2528, C :1267, B :1205 [for Sales Data]

Histogram of A,B,C and Stock-Turnover Ratio (For stock_data) [Image Name]

So we can clearly see the count of A,B,C is not same in stock data and sales_data means Items are moving from A to B to C , so I needed to Find those items which are frequently in A,B,C , moderate ,understocked and overstocked! Plotted a piecharts of sales count vs stock count of various types A,B,C respectively .So using the information of counts of A,B,C and



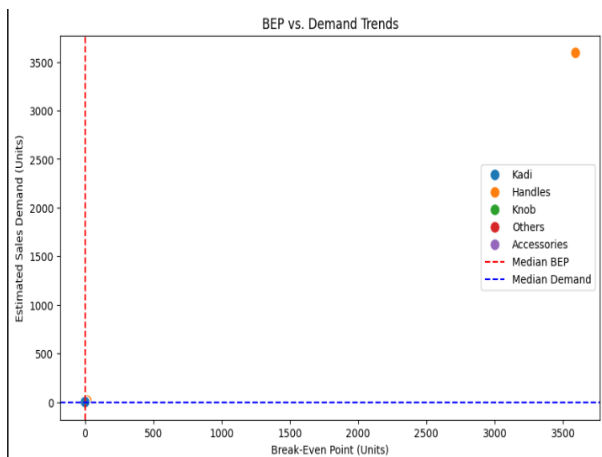
stock-turnover disparity we found-out which items come under which category and necessary action to be taken for that category which is.....

Categories Overlap and Items ,

Actions which can be taken to handle Problem [Table Name]

Category Overlap	Items	Actions To be Taken
Frequent-A & Frequent-C (16 items)	Antique Handle, KH-1239, Economy Kadi, Sofa Leg, Profile Handle, Tower Bolt, Khutti, Brass Khutti, Designer Knob, Cross Knob, Gold Plated Handle, Main Door Handle, Traditional Handle, Ultra Modern Handle, Key Stand, Kadi Deluxe	Ensure steady supply & improve demand forecasting to prevent fluctuations.
Frequent-C (19 items)	+3 More (Decorative Kadi, Stainless Steel Knob, Tower Bolt Premium)	Consider price adjustments or targeted promotions to push them into Frequent-A.
Understock & Frequent-A (12 items)	KH-1239, Economy Kadi, Profile Handle, Tower Bolt, Brass Khutti, Decorative Kadi, Cross Knob, Gold Plated Handle, Main Door Handle, Traditional Handle, Ultra Modern Handle, Key Stand	Urgently restock to avoid stockouts & lost sales.
Understock & Frequent-C (15 items)	Antique Handle, KH-1239, Economy Kadi, Sofa Leg, Profile Handle, Tower Bolt, Khutti, Brass Khutti, Decorative Kadi, Designer Knob, Cross Knob, Gold Plated Handle, Main Door Handle, Ultra Modern Handle, Key Stand	Urgently restock these fast-selling items to prevent missed revenue.
Overstock & Understock (1 item)	Tower Bolt	Review stock levels to prevent overstocking/understocking issues.
Frequent-B (3 items)	Decorative Kadi, Stainless Steel Knob, Tower Bolt Premium	Promote these items to increase sales turnover.

Image Name-> BEP VS Demand



5.1) Pareto Analysis – Delayed Orders by Category

The Pareto chart indicates that **Knobs and Handles** contribute the most to delayed orders, together accounting for a large cumulative percentage of delays. This highlights the **80/20 rule**, suggesting that focusing improvement efforts on these two categories could significantly reduce overall delays.

5.2) Monte Carlo Simulation – Machine Downtime Risk

The simulation reveals that **95% of machine downtime instances are under 21 hours**, with the distribution peaking around 12–15 hours. This quantifies the risk and emphasizes the importance of mitigating long-duration downtimes through preventive maintenance. Queuing Utilization Factor (ρ): 0.83 Avg Orders in Queue (L_q): 4.17 Avg Wait Time in Queue (W_q): 158.57 time units

5.3) Process Capability Analysis – Orders Processed

The histogram shows that most orders processed fall between **75 and 200**, with a clear **mean close to the center** of this range. However, there's considerable variability beyond upper and lower specification limits, indicating opportunities for **improving process stability** to maintain consistent throughput.

Additional Graphs for results as well as Descriptive Statistics in this link

https://drive.google.com/file/d/1Xoi2aWF2E3kT7sBolNgYY2_-JDNxxjG-/view?usp=sharing

After doing Analysis on sales and stock data we did analysis on finance_data and plotted scatter plot of

BEP vs Demand

Image Name-> Heat Map for showing overlap of Items in frequent-A, B, C, Understock, Overstock

